



WEEK 8: STATISTICS II – GRAPHICAL REPRESENTATION

TOPIC: Pie Charts; Interpretation of Data and Real-Life Applications

A. INTRODUCTION TO GRAPHICAL REPRESENTATION

Why We Use Graphs

Graphs help us to:

1. Present data visually for easy understanding
2. Compare values quickly
3. Show patterns and trends clearly
4. Communicate information effectively
5. Make data more attractive and meaningful

1. Pie Charts
2. Bar Charts
3. Line Graphs
4. Pictographs
5. Histograms

Types of Statistical Graphs

B. PIE CHARTS

Definition

A pie chart is a circular diagram divided into sectors (slices), where each sector represents a part of the whole.

Key Features

Whole circle = 360°

Each sector represents a proportion of the whole

Larger value $\rightarrow$ larger sector

Used for percentages and distribution

USE OF PIE CHARTS

When to Use Pie Charts

1. Showing parts of a whole
2. Small number of categories (2–8)



3. Percentage distribution
4. Budget analysis
5. Survey results
6. Basic Structure
7. Circle = Total data (100%)
8. Each slice = category share
9. Center = origin of all sectors

C. STEPS TO DRAW A PIE CHART

- Collect and organize data
- Find total frequency
- Calculate angles using the formula:

Formula for determining the angle of a sector of a pie chart :

$$\text{Angle of sector} = (\text{Frequency} / \text{Total Frequency}) \times 360^\circ$$

Steps:

- a) Draw a circle using a compass
- b) Draw a radius line
- c) Measure angles using a protractor
- d) Label each sector
- e) Add a title and legend



D. WORKED EXAMPLES

Example 1: Favorite Subjects

Subject	Frequency
Maths	8
English	6
Science	5
Social Studies	3

Step 1: calculate the Total

$$\text{Total} = 8 + 6 + 5 + 3 = 22$$

Step 2: calculate the Angles

- Maths: $(8/22) \times 360^\circ = 131^\circ$
- English: $(6/22) \times 360^\circ = 98^\circ$
- Science: $(5/22) \times 360^\circ = 82^\circ$
- Social Studies: $(3/22) \times 360^\circ = 49^\circ$
- Check: $131^\circ + 98^\circ + 82^\circ + 49^\circ = 360^\circ$ ✓

Example 2: Monthly Budget (₦60,000)

Item	Amount
1) Food	₦24,000
2) Rent	₦18,000
3) Transport	₦9,000
4) Others	₦9,000

Step 1: calculate the Total

$$\text{Total} = 24,000 + 18,000 + 9,000 + 9,000 = \text{₦60,000}$$

Step 2: calculate the Angle

$$\text{Food: } (24,000/60,000) \times 360^\circ = 144^\circ$$

$$\text{Rent: } (18,000/60,000) \times 360^\circ = 108^\circ$$

$$\text{Transport: } (9,000/60,000) \times 360^\circ = 54^\circ$$

$$\text{Others: } (9,000/60,000) \times 360^\circ = 54^\circ$$



Check: $144^\circ + 108^\circ + 54^\circ + 54^\circ = 360^\circ$ ✓

Example 3: Transport to School (40 students)

Mode	Frequency
Walk	12
Bus	15
Car	8
Bicycle	5

Step 1: Total = $12 + 15 + 8 + 5 = 40$

Step 2: Calculate the Angles

a. Walk: $(12/40) \times 360^\circ = 108^\circ$

b. Bus: $(15/40) \times 360^\circ = 135^\circ$

c. Car: $(8/40) \times 360^\circ = 72^\circ$

d. Bicycle: $(5/40) \times 360^\circ = 45^\circ$

E. INTERPRETING PIE CHARTS

To interpret:

Largest sector = most popular

Smallest sector = least popular

Convert angle to percentage

Compare categories

Percentage Formula

● Percentage = $(\text{Angle} / 360^\circ) \times 100$

Example: Apple = 144°

$(144/360) \times 100 = 40\%$

✓ Apple = 40% of total

F. REAL-LIFE APPLICATIONS

● Business



- Sales analysis
- Market share
- Personal Life
- Family budgeting
- Savings tracking
- Education
- Subject performance
- Time allocation
- Health
- Diet tracking
- Exercise monitoring

G. ADVANTAGES & LIMITATIONS

Advantages

- a. Easy to understand
- b. Visually clear
- c. Good for proportions
- d. Attractive presentation

Limitations

- Not suitable for many categories
- Not precise for comparisons
- Cannot show trends over time

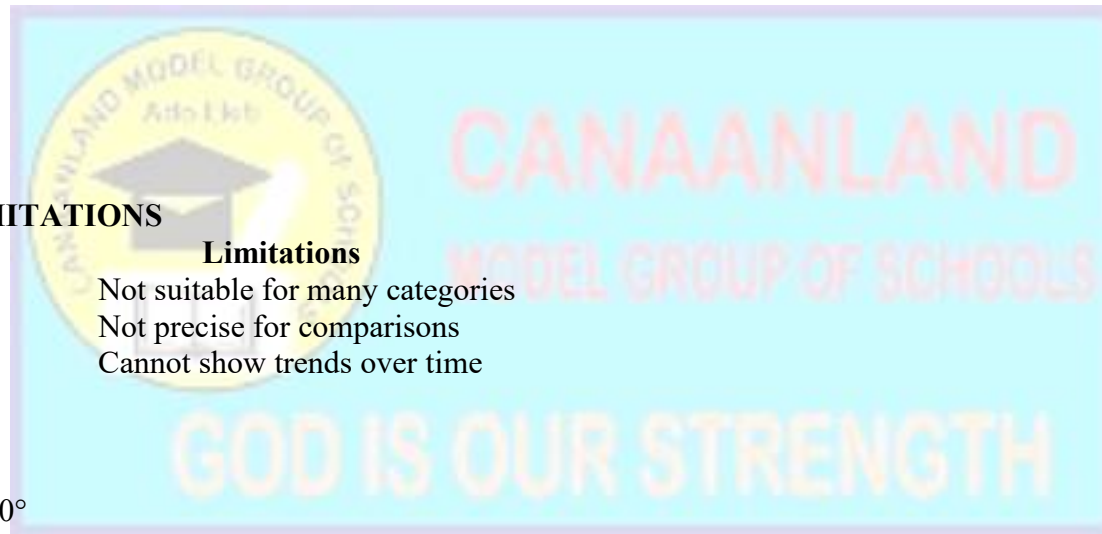
H. COMMON ERRORS

1. Angles not adding to 360°
2. Wrong calculations
3. Poor labelling
4. Incorrect protractor use

I. IMPORTANT EXAM EXAMPLE

Example: 60 Students Survey (Favorite Seasons)

Season	Frequency
Summer	24





Winter	18
Spring	12
Autumn	6

Step1: Find The Total = 60

Step 2: Find The Angles

- Summer: $(24/60) \times 360^\circ = 144^\circ$
- Winter: $(18/60) \times 360^\circ = 108^\circ$
- Spring: $(12/60) \times 360^\circ = 72^\circ$
- Autumn: $(6/60) \times 360^\circ = 36^\circ$

Check: $144^\circ + 108^\circ + 72^\circ + 36^\circ = 360^\circ$ ✓

INTERPRETATION

1. Most preferred: Summer
2. Least preferred: Autumn
3. Summer is 4 times Autumn

J. EVALUATION QUESTIONS

- I. Define a pie chart.
- II. What is the total angle in a pie chart?
- III. Why are pie charts used?
- IV. Calculate the angle for 15 out of 60.
- V. State 3 uses of pie charts.
- VI. What tool is used to measure angles?





VII. What does the largest sector mean?

VIII. Convert 120° to a percentage

IX. Draw a pie chart for given data.

X. List the limitations of pie charts.

K. REVISION QUESTIONS

1. What is a pie chart?
2. Why is 360° used in pie charts?
3. How do you compute angles for a pie chart?
4. What is the difference between a pie chart and a bar chart?
5. Can a pie chart show changes over time?

L. SUMMARY

Angle calculations

Pie chart drawing

Interpretation of data

Real-life applications

Word problems using percentages

